

Gajendra Saraswat

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PROFESSIONAL SUMMARY

Results-driven Software Engineer with **3+ years** (Aug 2023 – Present) combining **enterprise-scale backend** (Java, Spring Boot, Python) with hands-on **Machine Learning & AI** – fraud detection, medical imaging, and image compression. Built distributed systems using Java/Spring Boot and Python while deploying ML models with **Scikit-learn, TensorFlow, PyTorch**. Delivered **36% faster contract processing**, **99% faster portal loads**, **4,000+ vulnerabilities remediated**, and **95% test coverage** evolved to Jira/MCP automated regression. Published **Springer** compression research (5.5x–24.2x). Daily user of **Windsurf/Devin** for AI-assisted development and governance (RAG/Agent, prompt engineering via BNY AI Pioneer L3).

TECHNICAL SKILLS

- **Languages:** Python, Java (8, 17), C++, SQL
- **Data Science & AI:** Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, Matplotlib, Keras, OpenCV
- **LLM & AI Engineering:** RAG, Agent Design, Prompt Engineering, Nuggets, Workflow Orchestration, Windsurf, Devin, LLM-assisted Development
- **ML Engineering:** Model Development, Deployment, Predictive Modeling, Feature Engineering, Hyperparameter Tuning, Imbalanced Data (SMOTE), Evaluation (Precision/Recall)
- **Backend Development:** Spring Boot, RESTful APIs, Microservices, JPA/Hibernate, Flask, ASP.NET (legacy), API Gateway, API Manager, Monolithic & Distributed Systems
- **Databases & Storage:** MySQL, PostgreSQL, Redis, Vertica, DB2, Informatica (Regulatory Platform, ETL - platform-level), NoSQL, Database Design, Query Optimization
- **MLOps & Infrastructure:** Docker, Git, Jenkins, GitLab CI/CD, Maven, JFrog Artifactory, Apache Kafka, Ansible (Regulatory Platform), Google App Engine (Contract Platform), F5 BIG-IP, Cumulus, IIS, Windows Server 2016/2022
- **Observability & Quality:** Grafana, Splunk, Orion, PVT, JUnit, Mockito, Selenium, Cucumber, SonarQube, Jira, Confluence, MCP
- **Security:** Authentication, Authorization, RBAC/Entitlements, Secret Scanning, VMAD - Vulnerability Management, Site-Minder, CyberArk PAM

EXPERIENCE

Bank of New York Mellon

Software Engineer

Chennai, India

Aug 2023 – Present

- **Backend Development & ML Integration (Contract Platform – Java/Angular):** Enhanced Spring Boot microservices for contract processing with **@Async/@Transactional**, intelligent data validation, and anomaly detection concepts; optimized workflows -> **36% performance improvement**. Expose secure REST APIs and **Apache Kafka** async flows via **API Gateway / API Manager** (throttling, rate limiting, sandbox URLs); deploy Contract Platform via **Google App Engine**, Docker, GitLab CI/CD; strong fundamentals in distributed systems and scaling ML systems to production.
- **Legacy Modernization & Performance (Regulatory Platform – ASP.NET/IIS Monolithic):** Optimized legacy portal: **18.8 MB page 7 min -> 1 min (86% reduction)** via IIS static/dynamic compression enabled via RDP, then -> **5 sec (99% total reduction)** via **application-wide server-side pagination**. Demonstrates data-driven performance tuning and full-stack ownership from server to UI.
- **System Architecture, Disaster Recovery & Kafka Resiliency:** Architected DR workflows for **25+ applications** across multi-cluster; led Java 17/Angular 17 zero-downtime upgrades and migration of 30+ business-critical reports preserving entitlements. For resiliency testing, validated **Kafka backup cluster (8 topics, 2 clusters)** automatic failover (Cluster1 -> Cluster2), testing **earliest/latest** offsets and consumer lag in lower environments before production.
- **Infrastructure Migration (Regulatory Platform):** Migrated Windows Server **2016 -> 2022** (QA: 3 -> 2, PROD: 3 -> 3) for regulatory reporting platform (backed by DB2, Informatica ETL at platform level) via **Ansible** (Regulatory Platform) and **F5 BIG-IP** and **Cumulus** load balancer shifts (Cumulus URLs); validated per-server with isolated Chrome direct-URL testing; monitored via **Grafana, Splunk, Orion, PVT**; zero downtime, decommissioned old servers.
- **Security Hardening & Responsible AI Governance:** Spearheaded remediation of **4,000+ vulnerabilities** in 3 years (incl. **Regulatory Platform: 28 total – 5 Critical, 8 High**) securing login/auth flows (validated via SonarQube, VMAD - Vulnerability Management). Implemented RBAC (Approver/View/Add); **go-to person for entitlements** – create entitlements, map applications to level-based accesses, fulfill prod via **Access Hub** forms and directly provision lower environments. Audited **Devin Chain-of-Thought credential-exposure report** across chats, GitLab commits/pipelines; fixed exposures, rotated credentials, authored governance guidelines – applying **AI Pioneer L3** learnings on AI risk and compliance.
- **Data-Driven Insights, Quality & BAU:** Created Python scripts for system performance metrics, dashboards, and proactive anomaly detection. Built **NextGen automation (Selenium, Gherkin, Cucumber)** from **300+ tests / 95% coverage** to fully automated regression where scripts auto-generate cases/reports and push to **Jira via MCP (Model Context Protocol)**. Resolved P1s including **MIME outage (web.config text/html wrapping .axd/.css/.js fixed with SiteMinder)** and regulatory report before deadline. Own BAU/L2/L3, RFCs, runbooks.
- **AI-assisted Development:** Daily use of **Windsurf and Devin** for prompt-driven code generation, refactoring, debugging, test/docs drafting, and understanding unfamiliar stacks (e.g., ASP.NET/IIS); integrate **RAG/agent** patterns from AI Pioneer into backend and ML workflows.

SELECTED PROJECTS

- **Credit Card Fraud Detection ML System** *Sep 2020 – Final Year Project (ML Engineering)* [GitHub 16 stars/16 forks]
Designed ML-based fraud detection with RESTful APIs focusing on imbalanced data. Applied Logistic Regression with balanced class weights -> **97.6% accuracy, 89.6% recall**; handled imbalance via **SMOTE** and dimensionality reduction. Built Flask service with validation/error handling; containerized with **Docker** on **Heroku**. *Stack: Python, Scikit-learn, Flask, RESTful*

APIs, Docker, Heroku.

- **Leukemia Detection Using Deep Learning** *Jun 2019 – Sep 2019 – Internship Project (Computer Vision)* [GitHub: Blood-Cancer-Diagnosis-API] / hub.docker.com/r/sarques/bcpmodel (173 pulls)
Developed automated leukemia classification on microscopic cell images addressing morphological similarity. Implemented **SeRes-Next** via **PyTorch/Keras**, preprocessing with **Pandas/NumPy** -> **90% accuracy** via hyperparameter tuning. Containerized as **Docker** image on **Docker Hub** (173 pulls, [sarques/bcpmodel](https://hub.docker.com/r/sarques/bcpmodel)) for reproducible medical imaging pipeline.
- **Autoencoder-Huffman Ensemble for Image Compression** *Dec 2024 – Research Publication (ML & Data Compression)* [Springer Link]
Proposed novel two-stage autoencoder + Huffman encoding. Developed encoder-decoder networks for low-dimensional latent representations reducing noise while preserving features -> **5.5–24.2x compression** on MNIST, CIFAR-10, Fashion MNIST with minimal loss. Published in Springer proceedings. [AI-Without-Borders Projects]

EDUCATION

- **National Institute of Technology, Warangal** Warangal, India
Master of Technology, Computer Science; CGPA: 7.13 *Sep 2021 – Jun 2023*
- **MBM Engineering College, Jodhpur** Jodhpur, India
Bachelor of Engineering, Computer Science; CGPA: 7.48 *Aug 2016 – Oct 2020*

PUBLICATION

- **Autoencoder-Huffman Ensemble Model for Image Compression** *Springer, Singapore – Dec 2024 – Thesis* [Source]
Autoencoders compress via encoder-decoder creating low-dimensional latent representations (LSR); ensemble with Huffman achieves **5.5–24.2x** on MNIST/CIFAR-10/Fashion MNIST with minimal information loss.

HONORS & AWARDS

- **GATE 2021 Exam:** Secured **99.1 percentile (AIR 943)** in Graduate Aptitude Test in Engineering (Computer Science)
- **Kaggle SIIM-ISIC Melanoma Classification:** Ranked in **top 19% globally** in large-scale medical imaging competition

CERTIFICATIONS

- **AI Pioneer – Proficiency Level 3:** BNY Internal AI Learning Program – Prompt Engineering, RAG/Agent Design, Nuggets, Workflow Orchestration, AI Governance
- **Machine Learning** – Coursera
- **Deep Learning Specialization** – Coursera
- **Introduction to Data Science in Python** – Coursera
- **Statistical Thinking in Python (Part 1)** – DataCamp
- **Mathematics for Machine Learning** – Coursera

Keywords for ATS (visible): Python, Scikit-learn, TensorFlow, PyTorch, RAG, Agents, Prompt Engineering, Windsurf, Devin, Flask, Spring Boot, Microservices, Apache Kafka, Docker, Vertica, Grafana, Splunk, MCP, Jira